

Chapter Summary

Key Terms

7.1 The Multiplication Rule for Counting

- combinatorics
- Multiplication Rule for Counting (Fundamental Counting Principle)

7.2 Permutations

- permutation
- factorial

7.3 Combinations

- combination

7.4 Tree Diagrams, Tables, and Outcomes

- experiment
- replication
- sample space
- independent/dependent

7.5 Basic Concepts of Probability

- event
- probability
- theoretical probability
- empirical probability
- subjective probability

7.7 What Are the Odds?

- odds (for/against)

7.8 The Addition Rule for Probability

- mutually exclusive

7.9 Conditional Probability and the Multiplication Rule

- conditional probability

7.10 The Binomial Distribution

- binomial experiment
- probability density function (PDF)
- cumulative distribution function (CDF)

7.11 Expected Value

- expected value

Key Concepts

7.1 The Multiplication Rule for Counting

- The Multiplication Rule for Counting is used to count large sets.

7.2 Permutations

- Using the Multiplication Rule for Counting to enumerate permutations.
- Simplifying and computing expressions involving factorials.
- Using factorials to count permutations.

7.3 Combinations

- Permutations are used to count subsets when order matters; combinations work when order doesn't matter.
- Combinations can also be computed using factorials.

7.4 Tree Diagrams, Tables, and Outcomes

- We identify the sample space of an experiment by identifying all of its possible outcomes.
- Tables can help us find a sample space by keeping the possible outcomes organized.
- Tree diagrams provide a visualization of the sample space of an experiment that involves multiple stages.

7.5 Basic Concepts of Probability

- The theoretical probability of an event is the ratio of the number of equally likely outcomes in the event to the number of equally likely outcomes in the sample space.
- Empirical probabilities are computed by repeating the experiment many times, and then dividing the number of replications that result in the event of interest by the total number of replications.
- Subjective probabilities are assigned based on subjective criteria, usually because the experiment can't be repeated and the outcomes in the sample space are not equally likely.
- The probability of the complement of an event is found by subtracting the probability of the event from one.

7.6 Probability with Permutations and Combinations

- We use permutations and combinations to count the number of equally likely outcomes in an event and in a sample space, which allows us to compute theoretical probabilities.

7.7 What Are the Odds?

- Odds are computed as the ratio of the probability of an event to the probability of its complement.

7.8 The Addition Rule for Probability

- The Addition Rule is used to find the probability that one event or another will occur when those events are mutually exclusive.
- The Inclusion/Exclusion Principle is used to find probabilities when events are not mutually exclusive.

7.9 Conditional Probability and the Multiplication Rule

- Conditional probabilities are computed under the assumption that the condition has already occurred.
- The Multiplication Rule for Probability is used to find the probability that two events occur in sequence.

7.10 The Binomial Distribution

- Binomial experiments result when we count the number of successful outcomes in a fixed number of repeated, independent trials with a constant probability of success.
- The binomial distribution is used to find probabilities associated with binomial experiments.
- Probability density functions (PDFs) describe the probabilities of individual outcomes in an experiment; cumulative distribution functions (CDFs) give the probabilities of ranges of outcomes.

7.11 Expected Value

- The expected value of an experiment is the sum of the products of the numerical outcomes of an experiment with their corresponding probabilities.
- The expected value of an experiment is the most likely value of the average of a large number of replications of the experiment.

Formula Review

7.2 Permutations

$$\bullet \quad {}_n P_r = \frac{n!}{(n-r)!}$$

7.3 Combinations

- The formula for counting combinations is:

$${}_n C_r = \frac{n!}{r!(n-r)!}$$

7.5 Basic Concepts of Probability

- For an experiment whose sample space S consists of equally likely outcomes, the *theoretical probability* of the event E is the ratio $P(E) = \frac{n(E)}{n(S)}$ where $n(E)$ and $n(S)$ denote the number of outcomes in the event and in the

sample space, respectively.

- $P(E) = 1 - P(E')$

7.7 What Are the Odds?

- For an event E ,

$$\text{odds for } E = n(E) : n(E') = P(E) : P(E') = P(E) : (1 - P(E))$$

$$\text{odds against } E = n(E') : n(E) = P(E') : P(E) = (1 - P(E)) : P(E)$$

- If the odds in favor of E are $A : B$, then $P(E) = \frac{A}{A+B}$.

7.8 The Addition Rule for Probability

- If E and F are mutually exclusive events, then

$$P(E \text{ or } F) = P(E) + P(F).$$

- If E and F are events that contain outcomes of a single experiment, then

$$P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F).$$

7.9 Conditional Probability and the Multiplication Rule

- If E and F are events associated with the first and second stages of an experiment, then

$$P(E \text{ and } F) = P(E) \times P(F|E).$$

7.10 The Binomial Distribution

- Suppose we have a binomial experiment with n trials and the probability of success in each trial is p . Then:

$$P(\text{number of successes is } a) = {}_n C_a \times p^a \times (1 - p)^{n-a}$$

7.11 Expected Value

- If O represents an outcome of an experiment and $n(O)$ represents the value of that outcome, then the expected value of the experiment is:

$$\sum n(O) \times P(O)$$

where Σ stands for the sum, meaning we add up the results of the formula that follows over all possible outcomes.

Projects

1. The Binomial Distribution is one of many examples of a *discrete probability distribution*. Other examples include the Geometric, Hypergeometric, Multinomial, Poisson, and Negative Binomial Distributions. Choose one of these distributions, and find out what makes it different from the Binomial Distribution. In what situations can it be applied? How is it used? Once you have an idea of how it's used, write a series of five questions like the ones in this chapter that can be answered with that distribution, and find the answers.
2. *Binomial* is a word that also comes up in algebra; the word describes polynomials with two terms. At first glance, there isn't much to indicate that these two uses of the word are related, but it turns out there is a connection. Explore the connection between the Binomial Distribution and the algebraic concept of binomial expansion, (the process of multiplying out expressions like $(x + y)^n$ for a positive whole number n). Search for a connection with the mathematical object known as Pascal's Triangle.
3. Hazard is a dice game that was mentioned in Chaucer's *Canterbury Tales*. It was a popular game of chance played in taverns and coffee houses well into the 18th century; its popularity at the time of the foundation of probability theory means that it was a common example in early texts on finding expected values and probabilities. Find the rules of the game, and get some practice playing it. Then, analyze the choices that the caster gets to make, and decide which is most advantageous, using the language of expected values.

Chapter Review

The Multiplication Rule for Counting

1. You are booking a round trip flight for vacation. If there are 4 outbound flight options and 7 return flight options, how many different options do you have?
2. You are putting together a social committee for your club. You'd like broad representation, so you will choose one person from each class. If there are 8 seniors, 12 juniors, 10 sophomores, and 6 first-years, how many committees are possible?
3. The Big Breakfast Platter at Jimbo's Sausage Haus gives you your choice of 4 flavors of sausage, 5 preparations for eggs, 3 different potato options, and 4 different breads. If you choose one of each, how many different Big Breakfast Platters can be selected?
4. The multiple-choice quiz you're about to take has 10 questions with 4 choices for each. How many ways are there to fill out the quiz?

Permutations

5. Compute $\frac{8!}{2!3!3!}$.
6. Compute $\frac{12!}{8!3!}$.
7. Compute $\frac{211!}{210!}$.
8. Compute ${}_5P_3$.
9. Compute ${}_{15}P_3$.
10. Compute ${}_{22}P_5$.
11. As you plan your day, you see that you have 6 tasks on your to-do list. You'll only have time for 5 of those. How many schedules are possible for you today?
12. As captain of your intramural softball team, you are responsible for setting the 10-person batting order for the team. If there are 12 people on the team, how many batting orders are possible?

Combinations

13. If you're trying to decide which 4 of your 12 friends to invite to your apartment for a dinner party, are you using permutations or combinations?
14. If you're trying to decide which of your guests sits where at your table, are you using permutations or combinations?
15. Compute ${}_7C_4$.
16. Compute ${}_{13}C_8$.
17. How many ways are there to draw a hand of 8 cards from a deck of 16 cards?
18. In a card game with 4 players and a deck of 12 cards, how many ways are there to deal out the four 3-card hands?

Tree Diagrams, Tables, and Outcomes

19. If you draw a card at random from a standard 52-card deck and note its suit, what is the sample space?
20. If you draw 2 cards at random from a standard 52-card deck and note the 2 suits (without paying attention to the order), what is the sample space?
21. If you draw 2 *Scrabble* tiles in order without replacement from a bag containing E, E, L, S, what is the sample space?
22. If you draw 2 *Scrabble* tiles without replacement and ignoring order from a bag containing E, E, L, S, what is the sample space?
23. If you draw 2 *Scrabble* tiles without replacement and ignoring order from a bag containing E, E, L, S, what is the sample space?
24. If you draw 2 *Scrabble* tiles with replacement and ignoring order from a bag containing E, E, L, S, what is the sample space?

Basic Concepts of Probability

25. If you read that the probability of flipping 10 heads in a row is $\frac{1}{1024}$, is that probability most likely theoretical, empirical, or subjective?
26. If someone tells you that there is a 40% chance that a Democrat wins the U.S. Presidential election in 2132, is that probability most likely theoretical, empirical, or subjective?
27. If your professor says that you have a 20% chance of getting an A in her class because 20% of her students historically have earned As, is that probability most likely theoretical, empirical, or subjective?

In the following exercises, you are about to roll a standard 12-sided die (with faces labeled 1–12).

28. What is the probability of rolling a negative number?
29. What is the probability of rolling a number less than 20?
30. What is the probability of rolling an 11?
31. What is the probability of rolling a number less than 7?
32. What is the probability of *not* rolling an 11?
33. What is the probability of rolling a multiple of 4?
34. Over the last 30 years, it has rained 12 times on May 1. What empirical probability would you assign to the event "it rains next May 1"?

Probability with Permutations and Combinations

In the following exercises, you're drawing cards from a special deck of cards containing 2♥, 2♣, 2♦, 2♠, 3♥, 3♣, 3♠, 4♥, 4♣, 5♥.

35. If you draw 4 cards without replacement, what is the probability of drawing a 2, 3, 4, and 5 in order?
36. If you draw 4 cards without replacement, what is the probability of drawing a 2, 3, 4, and 5 in *any* order?
37. If you draw 3 cards without replacement, what is the probability that you draw a ♥, a ♣, and a ♥, in order?
38. If you draw 3 cards without replacement, what is the probability that you draw 2 ♥ and 1 ♣, in any order?

What Are the Odds?

39. If you roll a standard 20-sided die (with faces numbered 1–20), what are the odds against rolling a number less than 5?
40. If you roll a standard 20-sided die (with faces numbered 1–20), what are the odds in favor of rolling greater than a 5?
41. If $P(E) = \frac{4}{7}$, what are the odds in favor of E ?
42. If $P(E) = \frac{5}{17}$, what are the odds against E ?

The Addition Rule for Probability

In the following exercises, you're drawing a single card from a special deck of cards containing 2♥, 2♣, 2♦, 2♠, 3♥, 3♣, 3♠, 4♥, 4♣, 5♥.

43. What is the probability of drawing a 2 or a 3?
44. What is the probability of drawing a ♣ or a ♠?
45. What is the probability of drawing a 2 or a ♥?
46. What is the probability of drawing an even number or a ♣?

Conditional Probability and the Multiplication Rule

In the following exercises, you're drawing from a special deck of cards containing 2♥, 2♣, 2♦, 2♠, 3♥, 3♣, 3♠, 4♥, 4♣, 5♥.

47. If you draw a single card, what is:
- $P(\text{draw a 2})$
 - $P(\text{draw a 2} | \text{draw a ♥})$
 - $P(\text{draw a 2} | \text{draw a ♠})$
48. If you draw a single card, what is:
- $P(\text{draw a ♥})$
 - $P(\text{draw a ♥} | \text{draw a 3})$
 - $P(\text{draw a ♥} | \text{draw a 2})$

In the following exercises, you are playing the following game that involves rolling 2 dice, one at a time. First, you roll a

standard 6-sided die. If the result is a 4 or less, your second roll uses a standard 4-sided die. If the result of the first roll is a 5 or 6, your second roll uses a standard 6-sided die. Find these probabilities:

49. $P(\text{first roll is a 3 and second roll is a 3})$
50. $P(\text{first roll is a 6 and second roll is a 6})$
51. $P(\text{second roll is a 6})$
52. $P(\text{second roll is a 1})$

The Binomial Distribution

In the following exercises, decide whether the described experiment is a binomial experiment. If it is, identify the number of trials and the probability of success in each trial. If it isn't, explain why it isn't.

53. Draw 5 cards with replacement from a standard deck and count the number of ♠.
54. Draw 5 cards without replacement from a standard deck and count the number of ♠.
55. Draw cards from a standard deck and count how many cards are chosen before the first ♠ appears.

In the following exercises, you are about to roll a standard 20-sided die. Round answers to 4 decimal places.

56. Suppose you are going to roll the die 4 times. Give a full PDF table for the number of times a number greater than 16 appears.
57. If you roll the die 10 times, what is the probability that a number between 1 and 5 (inclusive) comes up exactly once?
58. If you roll the die 40 times, what is the probability that 20 comes up fewer than 2 times?
59. If you roll the die 40 times, what is the probability that 20 comes up 4 or more times?
60. If you roll the die 100 times, what is the probability that the number of times the die lands on something less than or equal to 7 is between 30 and 35 (inclusive)?
61. If you roll the die 100 times, what is the probability that the number of times the die lands on something less than or equal to 7 is exactly 36?
62. If you roll the die 100 times, what is the probability that the die lands on 20 between 5 and 8 times, inclusive?

Expected Value

63. You are playing a game where you roll a pair of standard 6-sided dice. You win \$32 if you get a sum of 12, and lose \$1 otherwise. What is the expected value of this game?
64. Interpret your answer.

You are playing a game where you roll a standard 12-sided die 4 times. If you roll 12 four times, you win \$1,000. If you roll 12 three times, you win \$100. If you roll 12 twice, you win \$10. If you roll 12 one time, you don't win or lose anything. If you roll don't roll a single 12, you lose \$1.

65. What is the expected value of this game?
66. Interpret your answer.

67. Which game would be better to play? Why?

Chapter Test

Each of the following exercises involve drawing a *Scrabble* tile from a bag. These tiles are labeled with a letter and a point value, as follows: A(1), C(3), D(2), E(1), E(1), J(8), K(5), O(1), R(1), R(1).

1. How many ways are there to draw a vowel and then a consonant from the bag?
2. How many ways are there to draw a tile worth an even number of points and then a tile worth an odd number of points from the bag?
3. How many ways are there to draw 4 tiles from the bag without replacement, if order matters?
4. How many ways are there to draw 4 consonants from the bag without replacement, if order matter?
5. How many ways are there to draw 4 tiles from the bag with replacement, if order does not matter?
6. How many ways are there to draw 4 consonants from the bag with replacement, if order does not matter?
7. Give the sample space of the experiment that asks you to draw 2 tiles from the bag with replacement and note their point values, where order doesn't matter. Give the outcomes as ordered pairs.
8. Give the sample space of the experiment that asks you to draw 2 tiles from the bag without replacement and note their point values, where order doesn't matter. Give the outcomes as ordered pairs.
9. If you draw a single tile from the bag, what is the probability that it's an E?
10. If you draw a single tile from the bag, what is the probability that it's *not* an A?
11. If you draw 3 tiles from the bag without replacement, what is the probability that they spell RED, in order?
12. If you draw 3 tiles from the bag without replacement, what is the probability that they spell RED, in *any* order?
13. What are the odds against drawing a vowel?
14. Use your answer to question 12 to find the odds against drawing three tiles without replacement and being

able to spell RED.

15. If you draw one tile, what is the probability of drawing a J or a K?
16. If you draw one tile, what is the probability that it's a vowel or that it's worth more than 4 points?
17. Suppose you're about to draw one tile from the bag. Find $P(\text{the letter is R})$ and $P(\text{the letter is R} | \text{the point value is 1})$.
18. If you draw 2 tiles with replacement, what is the probability of drawing a consonant first and then a vowel?
19. If you draw 2 tiles *without* replacement, what is the probability of drawing a consonant first and then a vowel?
20. If you draw 10 tiles with replacement, what is the probability that you draw exactly 3 vowels? Round to 3 decimal places.
21. If you draw 100 tiles with replacement, what is the probability that you draw fewer than 35 vowels? Round to 4 decimal places.
22. Find and interpret the expected number of points on the tile, assuming you draw 1 tile from the bag.
23. Find the expected sum of points on 2 tiles, selected without replacement.
24. If your friend offers you a bet where they pay you \$10 if you draw a vowel from the bag, but you owe them \$5 if you draw a consonant, should you take it? How do you know?